Days to the event from one full day – LSTM and transformer

Elazığ earthquake, 24 January 2020, 17:55 UTC (Mw 6.8) · stations FRT, KOVA, MAYA, NARI, SVRC · N component · generated 14 November 2026

1. The task

Question. Can the time that is left until the mainshock be read from one day of a station's recording? If the ground motion changed in some systematic way as the earthquake came closer, a model given a day it has not seen should place it at roughly the right distance from the event.

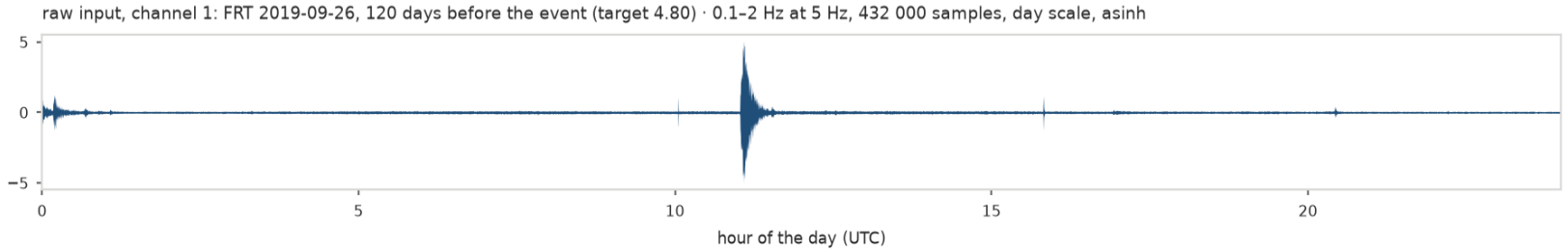
Sample and target. One sample is one station-day: the N component of one station from 00:00 to 24:00 UTC. The model gives one number for it, and the target is log(1 + d), where d is the number of calendar days from that day to the day of the mainshock (24 January 2020; d = 0 on that day, whose recording is cut at 17:00 UTC so that the mainshock itself is never in the input). The logarithm makes an error of a few days count as much close to the event as an error of months far from it: the days 0, 7, 30, 90 and 365 have the targets 0, 2.08, 3.43, 4.51 and 5.90. A model sees a single day at a time and has no information on the date or on the neighbouring days.

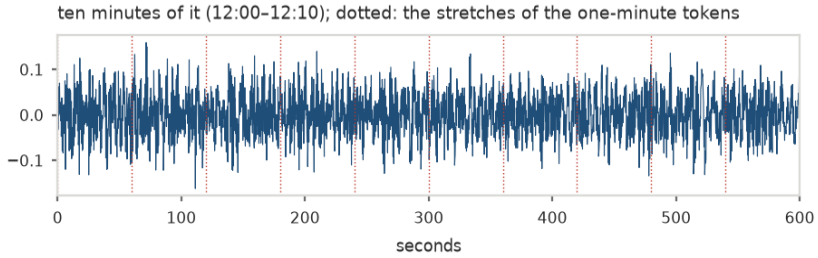
Four inputs for the same day. Each is a table of channels × time steps, and its last channel is a 0/1 mask that is 0 where data are missing.

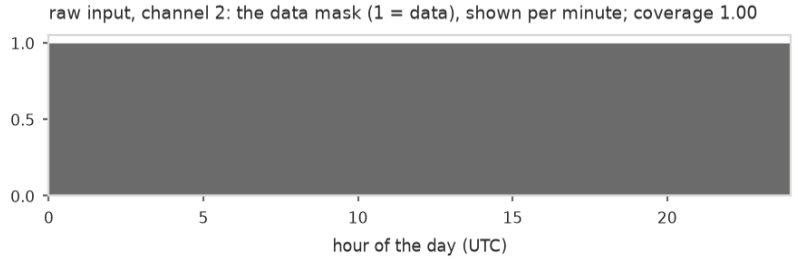
Input	Shape per day	What it holds	Scaling
raw, station scale	2 × 432 000	The signal band-passed to 0.1–2 Hz and resampled to 5 Hz (as in seismic-features-classification); gaps shorter than 2 s are interpolated, longer ones stay missing.	Divided by the station's median daily standard deviation of the training days, then asinh. A loud day stays loud in the input.
raw, day scale	2 × 432 000	The same signal.	Divided by the day's own standard deviation, then asinh. The level of the day is removed; only the shape of the signal is left.
features	13 × 432	The 12 features of seismic-features-classification for each of the day's 432 windows of 200 s (statistics of the window, mean frequency and period, two Lyapunov exponents, sample entropy, correlation dimension).	Per station: robust scaling fitted on the training windows, clipped at ±5. A window with a missing feature is zero with mask 0.
pca	8 × 432	The first 7 principal components of the scaled features.	PCA fitted on the training windows of the station.

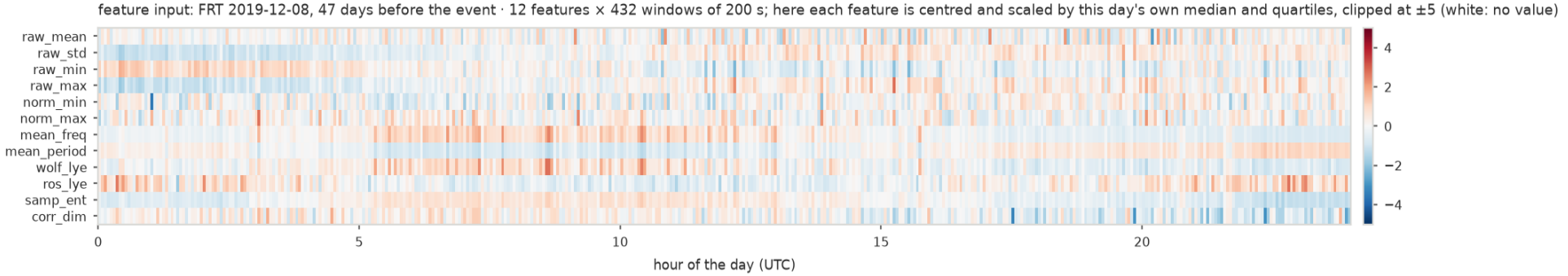
A station-day is used when at least 50 % of it has data (raw) or complete windows (features). The feature tables hold fewer days than the raw cache (section 6), so the feature inputs and the raw inputs are trained and scored on different sets of days.

One day as the models get it (a training day)





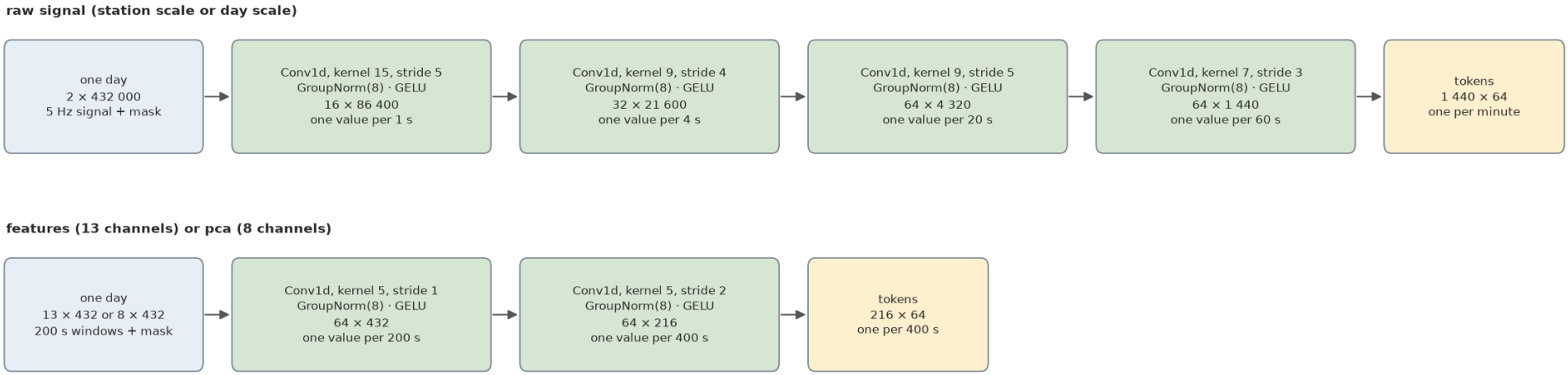




2. The models

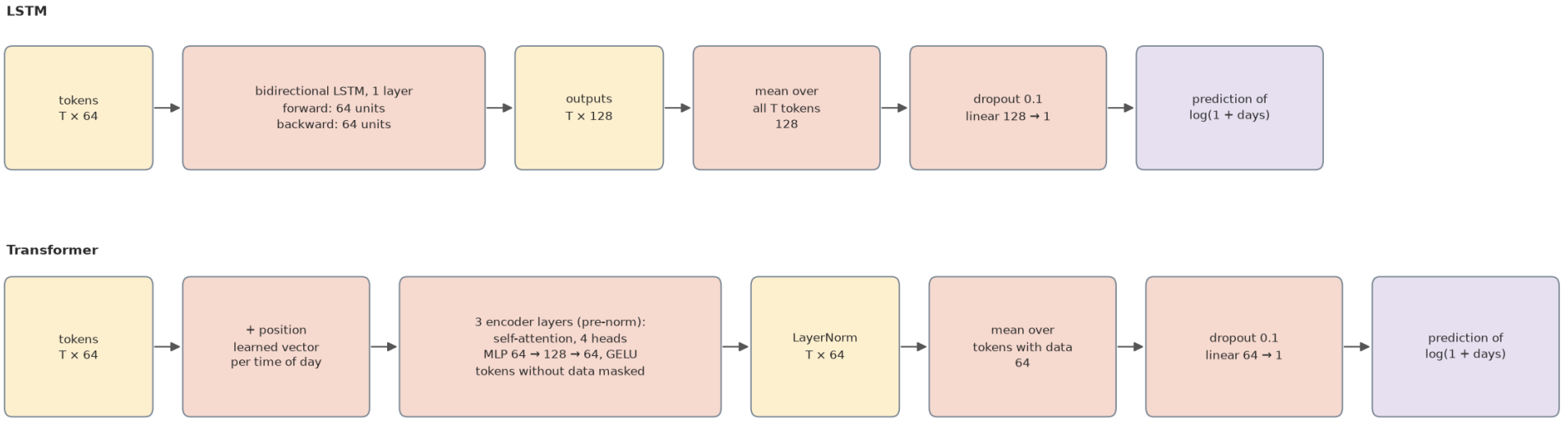
Both models work in two steps. A small convolutional network (the tokenizer) first shortens the day into a sequence of 64-dimensional vectors, the tokens: one per minute for the raw signal, one per 400 s for the feature inputs. A sequence model then reads all tokens of the day, and the average of its outputs is turned into the prediction by one linear layer. The two models share the tokenizer and the output layer and differ only in the sequence model.

Step 1: the CNN tokenizer. Each box is one layer with the shape of its output (channels × time steps).



The strides of the four raw layers multiply to 300, so 300 samples (60 s) become one token; through the widths of the kernels a token depends on 815 samples (163 s) around its minute. A feature token covers 2 windows and depends on 9 windows (30 min). The weights are the same at every position of the day.

Step 2: the sequence model and the output (T = 1 440 tokens for the raw signal, 216 for the feature inputs)



LSTM. A recurrent network reads the tokens one after the other and carries a memory of 64 numbers along; here one reads the day forwards and a second one backwards, and their outputs are put side by side, so every position has 128 numbers that depend on the whole day. Tokens from stretches without data are read like the others; the model can recognise them by the mask channel.

Transformer. Every token is compared with every other token of the day at once (self-attention), so a relation between two distant hours does not have to be carried through all the tokens between them. Since attention does not know the order of the tokens, a learned vector for the token's time of day is added first; it is the same on every day and so carries nothing about the date. Tokens whose stretch of the day has no data at all are left out of the attention and of the average.

Number of trained parameters

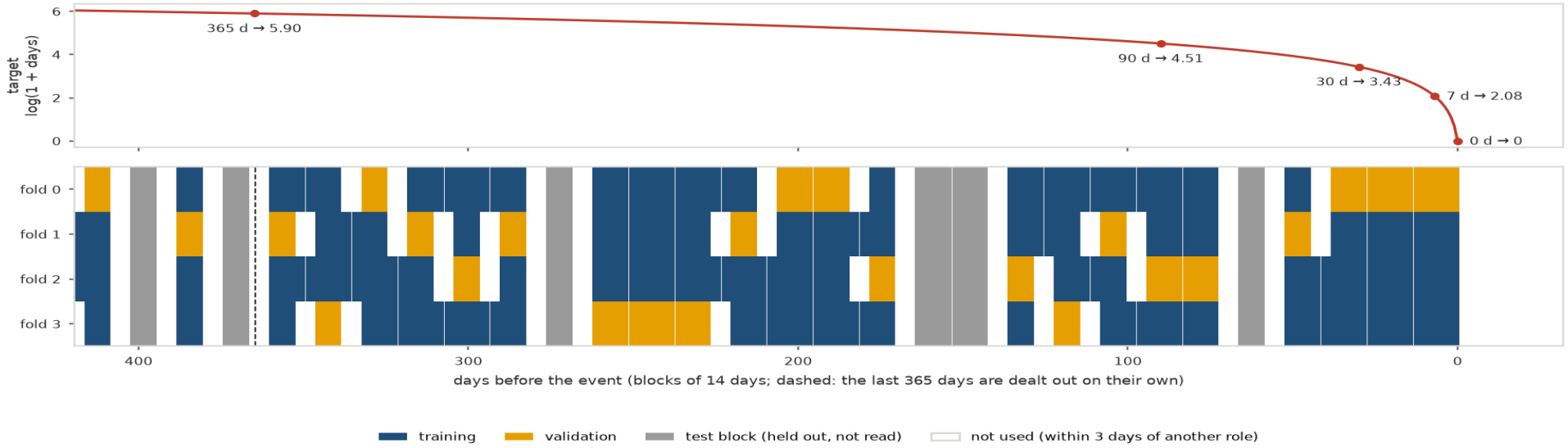
Input	Model	CNN tokenizer	Position table	Sequence model	Output layer	All parameters
raw (both scalings)	LSTM	52 720	–	66 560	129	**119 409**
raw (both scalings)	Transformer	52 720	92 160	100 544	65	**245 489**
features	LSTM	25 024	–	66 560	129	**91 713**
features	Transformer	25 024	13 824	100 544	65	**139 457**
pca	LSTM	23 424	–	66 560	129	**90 113**
pca	Transformer	23 424	13 824	100 544	65	**137 857**

The position table has one 64-dimensional vector per token, which is why the raw transformer is the largest model. Every model has far more parameters than a fold has training station-days (section 8).

3. Training and scoring

Training. Huber loss (δ = 1) on log(1 + d), AdamW, a one-cycle learning rate (rising over the first 10 % of the steps, then falling) with peak 1e-3 for the LSTM and 5e-4 for the transformer, weight decay 0.01 and 0.05, gradient norm clipped at 1, batches of 16 station-days, 40 epochs. The epoch with the lowest validation error is the one that is scored.

Split. Neighbouring days look alike, so days are not dealt out one by one. The calendar is cut into 14-day blocks counted back from the event, the same at every station, and a block is training, validation or test as a whole; a day within 3 days of a block of another role is not used. 20 % of the blocks are test blocks and were not read. The other blocks form 4 folds (the last 365 days and the earlier days dealt out separately): in fold k the blocks of fold k are validated and the rest are trained on, so over the 4 folds every one of these days is predicted once by a model that did not train on its block (out-of-fold predictions).



Runs. 4 inputs × 2 models × 3 seeds × 4 folds = 96 runs; the seed changes the initial weights and the order of the days, not the split.

Scores. MAE of log(1 + d) (lower is better; 0.69 is a factor of 2 in 1 + d) and the Spearman rank correlation between the true and the predicted time (1 = the days are put in the right order, 0 = no relation), over the out-of-fold predictions of a seed; given as mean ± standard deviation over the seeds, for all validation days and for the days of the last year.

Baselines, on the same validation days: the mean of the training targets (a model that learned nothing), and a straight-line fit per station on the day's level alone, i.e. the logarithm of its standard deviation (what can be had from how loud the day is, without anything else).

What a score can and cannot show. With one event, the time to the event is the calendar date counted backwards. A validation block lies between training blocks, so a model can score well by placing a day in the calendar from anything that changes slowly (season, noise level, the instrument), without any relation to the earthquake. A good score is therefore necessary for a precursor but does not show one; the pseudo-event placebo of section 9 asks how well the same models do for days on which nothing happened.

Results in short

Feature inputs, last year. MAE(log): features 1.29 ± 0.02 (LSTM) and 0.93 ± 0.04 (Transformer), pca 1.28 ± 0.04 (LSTM) and 0.93 ± 0.04 (Transformer); level-only baseline 1.36. Spearman: features 0.07 ± 0.04 (LSTM) and 0.63 ± 0.05 (Transformer), pca 0.02 ± 0.05 (LSTM) and 0.67 ± 0.06 (Transformer).

Raw signal, last year. MAE(log): station scale 0.88 ± 0.07 (LSTM) and 1.00 ± 0.07 (Transformer), day scale 0.91 ± 0.01 (LSTM) and 1.00 ± 0.05 (Transformer); level-only baseline 1.09. Spearman: station scale 0.29 ± 0.10 (LSTM) and 0.19 ± 0.06 (Transformer), day scale 0.29 ± 0.01 (LSTM) and 0.27 ± 0.06 (Transformer).

The final month. Median prediction for the days 0–30 before the event, in days: features 181 (LSTM) and 32 (transformer); pca 180 (LSTM) and 35 (transformer); raw, station scale 163 (LSTM) and 193 (transformer); raw, day scale 155 (LSTM) and 170 (transformer).

Transformer on the feature inputs, by kind of day. Median prediction in days, features and pca: final month 32 and 35; summer placebo block (178–207 days before) 34 and 40; spring placebo block 213 and 204; winter placebo block 33 and 75. Within the final month the days 0–14 and 15–30 get 32 and 32 days (features), 35 and 35 days (pca).

Where the final month is validated. All 153 validation station-days of the final month are in fold 0, together with the other days up to 41 days before the event; the closest training day of that fold is 45 days before the event (folds 1: 0, 2: 0, 3: 0 days). The models that predict the final month therefore had no training day from it. 74 of the 78 validation station-days of the summer placebo block are in the same fold.

Folds. Last-year Spearman of the single folds (predictions averaged over the seeds): features LSTM −0.40 to 0.55; features Transformer −0.54 to 0.58; pca LSTM −0.48 to 0.49; pca Transformer −0.28 to 0.64.

Stations, raw signal. Last-year Spearman at FRT: 0.53–0.67 in the four raw runs; at the other stations 0.07–0.29.

Pseudo-event placebo (section 9). Raw signal, 182 days of history per event, the earthquake and 13 pseudo-events. Rank of the earthquake among the 14 events (1 = best): raw, day scale LSTM: MAE 5, gain 5, Spearman 1, final-month AUC 6 (of 13); raw, day scale Transformer: MAE 4, gain 4, Spearman 2, final-month AUC 7 (of 13); raw, station scale LSTM: MAE 4, gain 1, Spearman 1, final-month AUC 2 (of 13); raw, station scale Transformer: MAE 3, gain 1, Spearman 1, final-month AUC 4 (of 13).

SVRC winters (section 10). With station scaling a countdown to 24 January transfers between winters whether or not they end with the earthquake. Spearman of models trained on one winter and predicting another: LSTM 0.29 to 0.31 between the two winters without an earthquake, 0.46 to 0.51 from the earthquake winter to those, 0.41 to 0.46 from those to the earthquake winter; Transformer 0.60 between the two winters without an earthquake, 0.62 from the earthquake winter to those, 0.68 to 0.73 from those to the earthquake winter. With day scaling: −0.03 to 0.26 over all pairs and both models.

Pseudo-event placebo at Balıkesir (section 11). Mainshock of 10 August 2025, raw signal, 84 days of history per event, the earthquake and 7 pseudo-events. Rank of the earthquake among the 8 events (1 = best): raw, day scale LSTM: MAE 8, gain 5, Spearman 1, final-month AUC 4 (of 8); raw, day scale Transformer: MAE 7, gain 5, Spearman 3, final-month AUC 7 (of 8); raw, station scale LSTM: MAE 8, gain 4, Spearman 1, final-month AUC 5 (of 8); raw, station scale Transformer: MAE 8, gain 5, Spearman 6, final-month AUC 7 (of 8).

With one event the days to the event are the calendar date counted backwards, and a validation block lies between training blocks, so a score here measures how well a day can be placed in the calendar from its recording; it does not by itself show a relation to the earthquake. The feature inputs and the raw inputs are scored on different days (the feature tables hold fewer days), so their rows are not comparable with each other.

4. Scores

Out-of-fold predictions of the validation days; mean ± standard deviation over 3 seeds. The baselines do not depend on the seed.

All validation days

Input	Station-days	MAE(log) train mean	MAE(log) level only	MAE(log) LSTM	MAE(log) transformer	Spearman level only	Spearman LSTM	Spearman transformer
features	490	1.53	1.47	**1.38 ± 0.01**	**1.13 ± 0.03**	−0.23	**0.00 ± 0.06**	**0.52 ± 0.03**
pca	490	1.53	1.47	**1.34 ± 0.03**	**1.08 ± 0.04**	−0.23	**−0.04 ± 0.04**	**0.60 ± 0.07**
raw, station scale	1448	1.10	1.05	**0.88 ± 0.07**	**0.90 ± 0.06**	0.16	**0.45 ± 0.14**	**0.47 ± 0.07**
raw, day scale	1448	1.10	1.05	**0.78 ± 0.01**	**0.84 ± 0.05**	0.16	**0.63 ± 0.01**	**0.57 ± 0.05**

Validation days of the last year (0–365 days before the event)

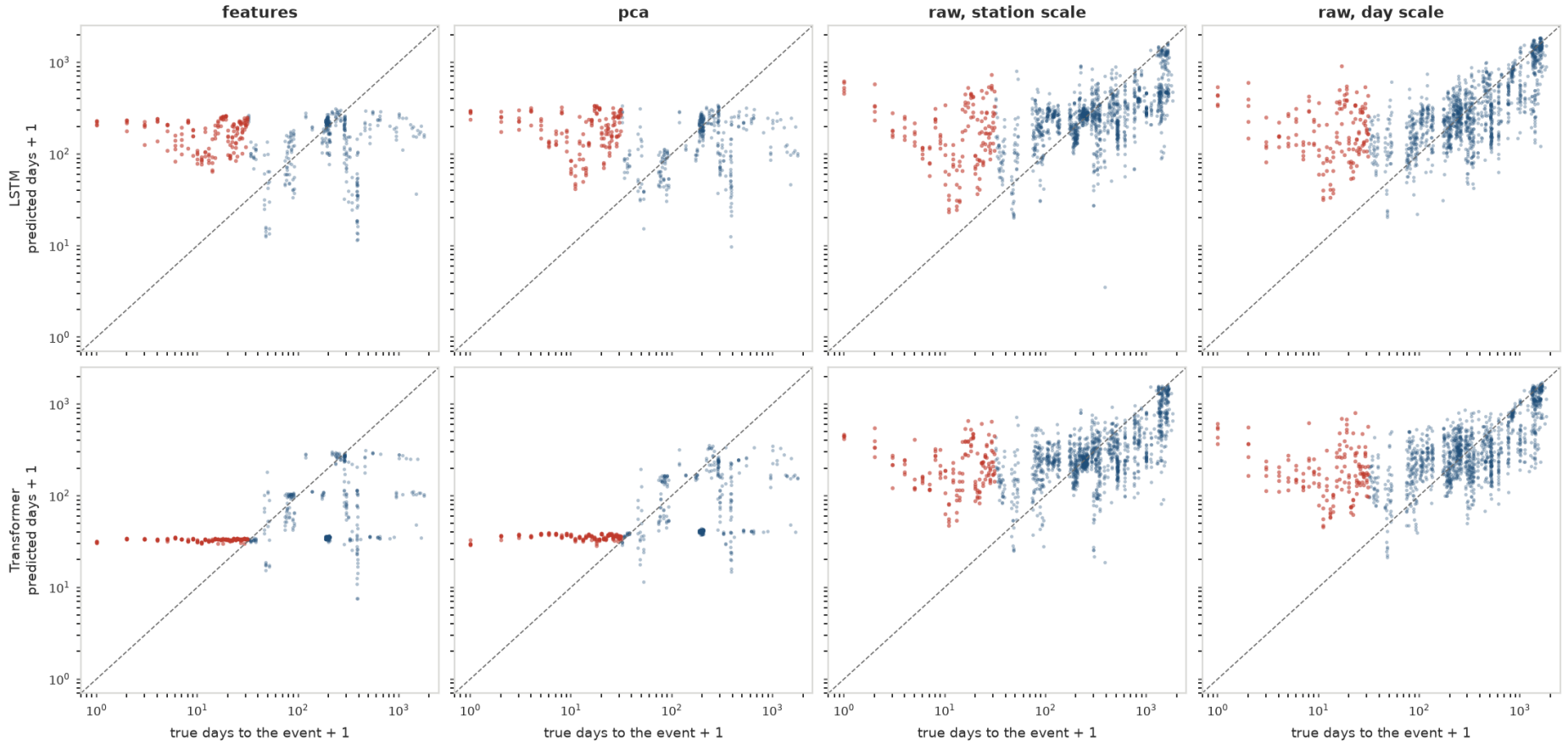
Input	Station-days	MAE(log) train mean	MAE(log) level only	MAE(log) LSTM	MAE(log) transformer	Spearman level only	Spearman LSTM	Spearman transformer
features	414	1.48	1.36	**1.29 ± 0.02**	**0.93 ± 0.04**	−0.13	**0.07 ± 0.04**	**0.63 ± 0.05**
pca	414	1.48	1.36	**1.28 ± 0.04**	**0.93 ± 0.04**	−0.13	**0.02 ± 0.05**	**0.67 ± 0.06**
raw, station scale	913	1.04	1.09	**0.88 ± 0.07**	**1.00 ± 0.07**	−0.53	**0.29 ± 0.10**	**0.19 ± 0.06**
raw, day scale	913	1.04	1.09	**0.91 ± 0.01**	**1.00 ± 0.05**	−0.53	**0.29 ± 0.01**	**0.27 ± 0.06**

Spearman per station, last year (predictions averaged over the seeds)

Input	Model	FRT	KOVA	MAYA	NARI	SVRC
features	LSTM	0.35	−0.00	0.13	0.04	−0.25
features	Transformer	0.43	0.69	0.75	0.63	0.60
features	level only · station-days	−0.65 · 82	0.21 · 94	−0.55 · 94	−0.44 · 94	0.13 · 50
pca	LSTM	0.12	0.07	0.15	−0.04	−0.45
pca	Transformer	0.70	0.69	0.79	0.69	0.61
pca	level only · station-days	−0.65 · 82	0.21 · 94	−0.55 · 94	−0.44 · 94	0.13 · 50
raw, station scale	LSTM	0.67	0.28	0.20	0.20	0.20
raw, station scale	Transformer	0.62	0.12	0.13	0.07	0.07
raw, station scale	level only · station-days	−0.55 · 208	−0.41 · 208	−0.55 · 208	−0.53 · 208	−0.52 · 81
raw, day scale	LSTM	0.53	0.27	0.22	0.23	0.09
raw, day scale	Transformer	0.63	0.24	0.29	0.16	0.09
raw, day scale	level only · station-days	−0.55 · 208	−0.41 · 208	−0.55 · 208	−0.53 · 208	−0.52 · 81

5. Predicted against true time

Every point is one validation station-day, predicted by the models that did not train on its block and averaged over the seeds. Both axes are logarithmic (days + 1); the dashed line is a correct prediction. Red: the final 30 days.



Predicted days per range of the true time: median (10th–90th percentile)

Input	Model	0–14 days	15–30 days	31–90 days	91–180 days	181–365 days	more than 365 days
features	LSTM	**156** (83–227)	**200** (101–251)	**93** (28–166)	**146** (52–252)	**206** (93–258)	**154** (18–254)
features	Transformer	**32** (30–33)	**32** (32–33)	**56** (30–103)	**103** (61–264)	**35** (33–273)	**77** (20–268)
features	station-days	75	78	85	19	157	76
pca	LSTM	**152** (62–292)	**208** (121–312)	**78** (39–142)	**97** (51–231)	**207** (91–281)	**126** (36–254)
pca	Transformer	**35** (32–37)	**35** (32–37)	**55** (30–160)	**162** (53–216)	**41** (39–246)	**114** (29–250)
pca	station-days	75	78	85	19	157	76
raw, station scale	LSTM	**125** (32–335)	**198** (75–466)	**143** (49–303)	**252** (164–309)	**257** (130–361)	**377** (164–1069)
raw, station scale	Transformer	**171** (84–391)	**230** (112–452)	**167** (75–333)	**244** (161–370)	**239** (131–396)	**431** (183–1306)
raw, station scale	station-days	75	78	169	156	435	535
raw, day scale	LSTM	**135** (51–363)	**184** (82–431)	**126** (67–243)	**218** (103–313)	**240** (118–429)	**570** (172–1463)
raw, day scale	Transformer	**153** (68–365)	**176** (94–480)	**144** (67–341)	**252** (147–422)	**260** (139–458)	**501** (200–1410)
raw, day scale	station-days	75	78	169	156	435	535

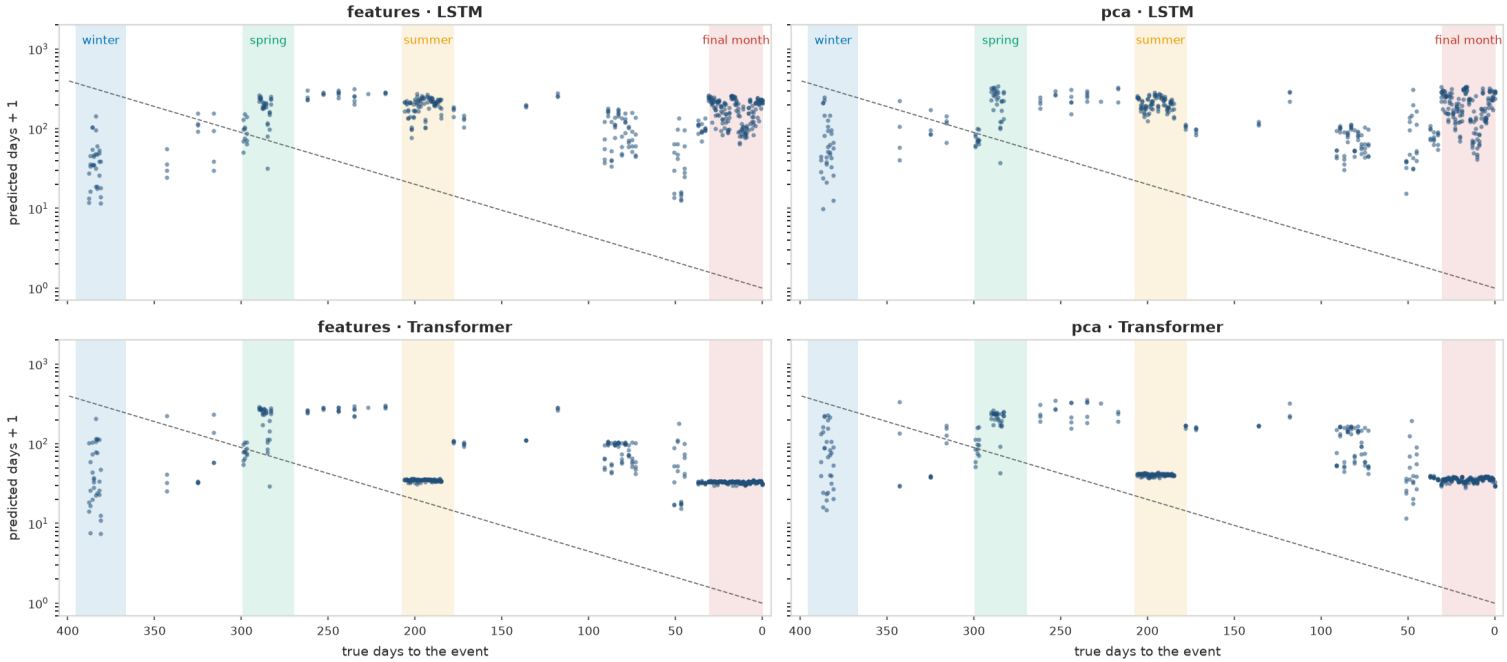
6. Feature inputs: predictions by kind of day

The feature tables do not hold every day. They hold every day of the final month, every 2nd day of days 31–90 and every 9th day of days 91–365 (the days the sfc run sampled), single days of the earlier years, and every day of the three 30-day placebo blocks (summer 178–207 days before the event, spring 270–299 days before the event, winter 367–395 days before the event; no placebo blocks at SVRC). Below: the validation days of each kind and what the models predict for them. Days in test blocks or in the gaps between blocks are not included.

Predicted days per kind of day: median (10th–90th percentile), predictions averaged over the seeds

Kind of day	Validation station-days	True days to the event	Fold: station-days	features · LSTM predicted days	features · Transformer predicted days	pca · LSTM predicted days	pca · Transformer predicted days
Final month, every day (0–30 days)	153	0–30	0: 153	**181** (91–239)	**32** (31–33)	**180** (77–301)	**35** (32–37)
Summer placebo block, every day	78	178–206	0: 74, 2: 4	**206** (133–239)	**34** (33–35)	**199** (146–257)	**40** (38–42)
Spring placebo block, every day	34	283–299	1: 26, 2: 8	**180** (69–235)	**213** (73–274)	**220** (66–302)	**204** (74–244)
Winter placebo block, every day	31	381–388	1: 31	**34** (13–60)	**33** (12–113)	**50** (23–208)	**75** (19–213)
Sampled days, 31–365 days (every 2nd or 9th day)	149	31–343	0: 32, 1: 31, 2: 61, 3: 25	**114** (35–264)	**73** (31–268)	**96** (43–267)	**89** (33–240)
Sampled days, more than a year before	45	384–1767	0: 14, 1: 12, 2: 13, 3: 6	**197** (147–276)	**105** (34–283)	**216** (98–265)	**161** (39–269)

The closest training day of fold 0 is 45 days before the event, so log(1 + days) of its training days is never below 3.83; the transformer's predictions for the final month and for the summer block lie below that floor.



7. The single folds

Last year only. Per fold: Spearman of the predictions averaged over the seeds, and below it the lowest and highest value of the single seeds. A fold's validation days are the days of its blocks, so the folds differ in how many days they hold and where in the year these lie. Closest training day, in days before the event: fold 0: 45, fold 1: 0, fold 2: 0, fold 3: 0.

Input	Model	Fold 0	Fold 1	Fold 2	Fold 3
features	LSTM	**0.13** −0.01 to 0.23	**0.55** 0.50 to 0.58	**0.21** 0.19 to 0.23	**−0.40** −0.56 to −0.28
features	Transformer	**0.51** 0.31 to 0.61	**0.58** 0.53 to 0.65	**0.24** 0.19 to 0.32	**−0.54** −0.68 to −0.32
features	station-days · of them in the final month	259 · 153	57 · 0	73 · 0	25 · 0
pca	LSTM	**0.04** −0.03 to 0.11	**0.49** 0.14 to 0.62	**0.27** 0.26 to 0.28	**−0.48** −0.50 to −0.48
pca	Transformer	**0.64** 0.35 to 0.64	**0.59** 0.55 to 0.62	**0.24** 0.17 to 0.28	**−0.28** −0.38 to −0.18
pca	station-days · of them in the final month	259 · 153	57 · 0	73 · 0	25 · 0
raw, station scale	LSTM	**0.10** −0.06 to 0.16	**0.36** 0.35 to 0.40	**0.17** −0.03 to 0.21	**−0.13** −0.34 to −0.13
raw, station scale	Transformer	**0.03** −0.02 to 0.06	**0.36** 0.28 to 0.40	**0.17** 0.04 to 0.22	**−0.04** −0.14 to 0.47
raw, station scale	station-days · of them in the final month	313 · 153	179 · 0	206 · 0	215 · 0
raw, day scale	LSTM	**0.13** 0.11 to 0.14	**0.41** 0.37 to 0.40	**0.29** 0.13 to 0.31	**−0.19** −0.20 to −0.11
raw, day scale	Transformer	**0.11** 0.01 to 0.14	**0.40** 0.27 to 0.41	**0.20** 0.07 to 0.27	**−0.10** −0.14 to 0.01
raw, day scale	station-days · of them in the final month	313 · 153	179 · 0	206 · 0	215 · 0

8. The runs

Input	Model	Parameters	Learning rate (peak)	Weight decay	Training station-days per fold	Epoch with the lowest validation error
features	LSTM	91 713	0.001	0.01	279–549	median 10 (3–24)
features	Transformer	139 457	0.0005	0.05	279–549	median 4 (2–20)
pca	LSTM	90 113	0.001	0.01	279–549	median 6 (3–17)
pca	Transformer	137 857	0.0005	0.05	279–549	median 5 (2–17)
raw, station scale	LSTM	119 409	0.001	0.01	1236–1493	median 28 (2–33)
raw, station scale	Transformer	245 489	0.0005	0.05	1236–1493	median 15 (2–32)
raw, day scale	LSTM	119 409	0.001	0.01	1236–1493	median 27 (13–34)
raw, day scale	Transformer	245 489	0.0005	0.05	1236–1493	median 19 (9–29)

Run on NVIDIA GeForce RTX 5090, torch 2.15.0+cu132, python 3.13.18. Commit recorded by the runs: 80e1a1a3c14c37988008ac8d7f94e3a7ace22a7c (uncommitted changes). The code was copied to the server before it was committed; its files were compared afterwards and are those of commit 25620fb. The epoch that is scored is chosen on the validation days themselves, which favours every model a little and both models in the same way.

9. Pseudo-event placebo

What was run. The same training with the days counted to another day: the day of the earthquake and 13 pseudo-events, 14, 28, … days before it (2019-07-26 to 2020-01-10), on which no earthquake happened. Every event gets only the 182 days up to it, so all events have the same length of history; the real event is run again under this limit and its scores here are not those of section 4. Raw signal only (the feature tables do not hold every day of 2019), stations FRT, KOVA, MAYA, NARI (SVRC has no data from February to September 2019), both scalings, both models, 4 folds, one seed: 224 runs with 284–456 training station-days each.

Split. The blocks are the calendar blocks of the main runs, and their test blocks are not read here either. For every event the other blocks are dealt into the folds in the order of their age (the block that ends at the event to fold 0, the next to fold 1, …), so the blocks closest to an event are validated by different folds and each has training days on its far side. This differs from the split of sections 4–8.

Scores per event (out-of-fold predictions of its validation days): MAE of log(1 + days); the gain, i.e. the MAE of the level-only baseline minus the model's MAE; the Spearman correlation per fold, averaged; and the final-month AUC, the share of pairs of a day 0–30 days before the event and an earlier day in which the closer day gets the shorter prediction (0.5 = chance). Rank 1 is the best of the events. With 14 events the smallest share a rank can give is 1/14 = 0.07; the events share most of their days and are not independent.

The real event among the pseudo-events

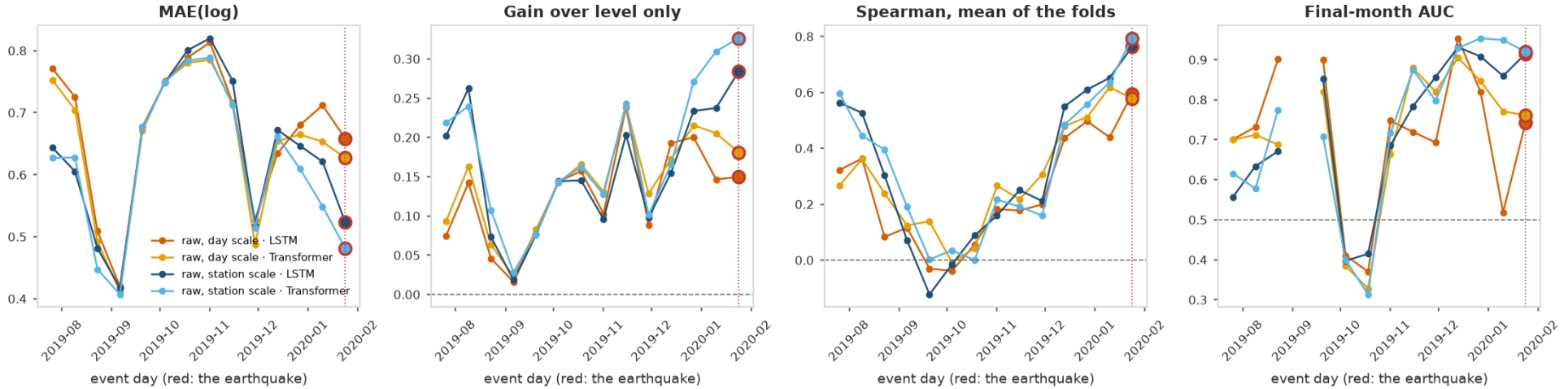
Input	Model	MAE(log) real · pseudo median (range) · rank of real	Gain over level only real · pseudo median (range) · rank of real	Spearman, mean of the folds real · pseudo median (range) · rank of real	Final-month AUC real · pseudo median (range) · rank of real
raw, station scale	LSTM	**0.52** · 0.65 (0.42 to 0.82) · **4 of 14**	**0.28** · 0.15 (0.02 to 0.26) · **1 of 14**	**0.76** · 0.25 (−0.12 to 0.65) · **1 of 14**	**0.92** · 0.74 (0.40 to 0.93) · **2 of 13**
raw, station scale	Transformer	**0.48** · 0.63 (0.41 to 0.79) · **3 of 14**	**0.33** · 0.16 (0.03 to 0.31) · **1 of 14**	**0.79** · 0.22 (0.00 to 0.64) · **1 of 14**	**0.92** · 0.75 (0.31 to 0.95) · **4 of 13**
raw, station scale	level only	**0.81** · 0.86 (0.44 to 0.95) · **5 of 14**	–	–	**0.63** · 0.24 (0.16 to 0.66) · **2 of 13**
raw, day scale	LSTM	**0.66** · 0.71 (0.42 to 0.81) · **5 of 14**	**0.15** · 0.14 (0.02 to 0.24) · **5 of 14**	**0.59** · 0.18 (−0.04 to 0.50) · **1 of 14**	**0.74** · 0.73 (0.37 to 0.95) · **6 of 13**
raw, day scale	Transformer	**0.63** · 0.67 (0.41 to 0.79) · **4 of 14**	**0.18** · 0.14 (0.02 to 0.24) · **4 of 14**	**0.58** · 0.27 (−0.01 to 0.62) · **2 of 14**	**0.76** · 0.74 (0.33 to 0.91) · **7 of 13**
raw, day scale	level only	**0.81** · 0.86 (0.44 to 0.95) · **5 of 14**	–	–	**0.63** · 0.24 (0.16 to 0.66) · **2 of 13**

Cheat note: how to read the table above

• A cell is: real · pseudo median (range) · rank. Real is the score with the countdown aimed at the earthquake, pseudo the same score for the 13 dates without one. Rank 1 is the best; 1 of 14 is the most a rank can say.

• MAE: lower is better. Gain above 0: better than loudness alone. Spearman of the folds: are the days put in the right order (1 perfect, 0 none); the score to trust most. Final-month AUC: 0.5 is chance.

• The level only rows are loudness alone: a straight line on the level of the day.



Every event: MAE(log) · final-month AUC

Event day	Days before the earthquake	Validation station-days	of them 0–30 days	Level only MAE · AUC	raw, station scale · LSTM MAE · AUC	raw, station scale · Transformer MAE · AUC	raw, day scale · LSTM MAE · AUC	raw, day scale · Transformer MAE · AUC
2020-01-24 (earthquake)	0	344	76	0.81 · 0.63	0.52 · 0.92	0.48 · 0.92	0.66 · 0.74	0.63 · 0.76
2020-01-10	14	344	76	0.86 · 0.66	0.62 · 0.86	0.55 · 0.95	0.71 · 0.52	0.65 · 0.77
2019-12-27	28	344	76	0.88 · 0.60	0.65 · 0.91	0.61 · 0.95	0.68 · 0.82	0.66 · 0.85
2019-12-13	42	344	44	0.83 · 0.16	0.67 · 0.93	0.66 · 0.93	0.63 · 0.95	0.65 · 0.91
2019-11-29	56	332	32	0.62 · 0.16	0.52 · 0.86	0.51 · 0.80	0.53 · 0.69	0.49 · 0.82
2019-11-15	70	376	76	0.95 · 0.27	0.75 · 0.78	0.71 · 0.87	0.72 · 0.72	0.71 · 0.88
2019-11-01	84	376	76	0.92 · 0.25	0.82 · 0.69	0.79 · 0.72	0.81 · 0.75	0.79 · 0.66
2019-10-18	98	332	76	0.95 · 0.38	0.80 · 0.41	0.78 · 0.31	0.79 · 0.37	0.78 · 0.33
2019-10-04	112	344	76	0.89 · 0.22	0.75 · 0.40	0.75 · 0.40	0.75 · 0.41	0.75 · 0.38
2019-09-20	126	344	44	0.75 · 0.21	0.68 · 0.85	0.68 · 0.71	0.68 · 0.90	0.67 · 0.82
2019-09-06	140	332	0	0.44 · –	0.42 · –	0.41 · –	0.42 · –	0.41 · –
2019-08-23	154	364	32	0.55 · 0.23	0.48 · 0.67	0.45 · 0.77	0.51 · 0.90	0.49 · 0.69
2019-08-09	168	408	76	0.87 · 0.21	0.61 · 0.63	0.63 · 0.58	0.73 · 0.73	0.70 · 0.71
2019-07-26	182	368	76	0.85 · 0.31	0.64 · 0.56	0.63 · 0.62	0.77 · 0.70	0.75 · 0.70

Notes. The station-days are those of the first row's input; both scalings use the same days. The events do not have the same validation days relative to themselves (columns 3 and 4), so the MAE of two events is not taken over the same range of times to the event. Events with fewer validation days in their last 30 days (2019-12-13, 2019-11-29, 2019-09-20, 2019-09-06, 2019-08-23) have a test block there, which is not read. The recording of the day of the earthquake is cut at 17:00 UTC, before the mainshock; the last day of a pseudo-event is a whole day. The epoch that is scored is chosen on the validation days, for every event in the same way. Commit recorded by the placebo runs: 25620fb0e8f3f77da816fcbc63a8b3c6729cf34c (uncommitted changes).

Cheat note: what the tables of this section show

• Spearman of the folds for the earthquake: 0.58 to 0.79 over the four setups; the highest of any pseudo-event: 0.65. For the two dates closest to the earthquake: 0.44 to 0.65 for 2020-01-10; 0.50 to 0.61 for 2019-12-27.

• Loudness alone, final-month AUC: 0.63 for the earthquake, median 0.24 for the pseudo-events.

• Table of every event: one line per target date, cells are MAE · final-month AUC. Compare a cell with the level only cell of its line, not with other lines: the lines differ in their validation days. A low MAE on a line with few days in the last 30 is a narrow range of targets, not a good prediction.

• Reading: with station scale the earthquake scores best, but the dates two and four weeks before it score almost as well, and loudness alone already separates its final month. The models follow something that changes slowly through December and January, not the day of the earthquake.

Cheat note: station scale and day scale

• Station scale: every day is divided by one number per station (the median standard deviation of its training days). A loud day stays loud, so the model sees how the day compares with the station's usual level.

• Day scale: every day is divided by its own standard deviation. The loudness of the day as a whole is removed; what is left is its shape (the course within the day, how spiky it is, its frequency content).

• A result that holds with day scale comes from the shape; one that holds only with station scale comes mainly from the loudness of the day. Day scale does not remove everything that goes along with a loud day.

10. SVRC: winters with and without the earthquake

Why. In section 9 the day of the earthquake and mid-winter coincide. SVRC is the one station with continuous recordings in other winters: 2015-16 and 2018-19, which did not end with an earthquake, and 2019-20, which did.

Transfer between winters. A model is trained on the 115 days up to 24 January of one winter (or of two) and predicts the days of another winter, every day counted to the 24 January of its own winter. Nothing of the predicted winter is used in training, and the last epoch is scored. If the time to the earthquake is read from the season, a model that has only seen a winter without an earthquake orders the days of the earthquake winter about as well as it orders those of another winter without one. Raw signal, both scalings, both models, 5 seeds; scores are the mean ± standard deviation over the seeds. Days in test blocks or within 3 days of one are not read. Usable days per winter: 2015-16 57 (of them 13 in the last 30 days, closest 0 days before 24 January); 2018-19 76 (of them 5 in the last 30 days, closest 19 days before 24 January); 2019-20 (earthquake) 92 (of them 29 in the last 30 days, closest 0 days before 24 January).

What this shows (raw, station scale · Transformer, trained on one winter, mean of the seeds).

• From one winter without an earthquake to the other: Spearman 0.60 (level only 0.42 to 0.43). A countdown to 24 January is learned where no earthquake followed.

• From the earthquake winter to the winters without: 0.62. What is learned from the earthquake winter works equally where no earthquake follows.

• From the winters without to the earthquake winter: 0.68 to 0.73, but level only already gives 0.63. The model adds 0.05 to 0.10 to level only here, 0.17 to 0.20 in the winters without an earthquake.

• Order, not timing: for the last 30 days before the earthquake the median prediction is 31 to 50 days.

• Day scale (the day's level removed), both models: −0.03 to 0.26. What transfers between winters is mainly the level of the day.

• Reading: the countdown at SVRC does not need an earthquake; season alone can produce it. Not ruled out: a precursor underneath the seasonal trend. One station, three winters, 57 to 92 usable days each.

Spearman correlation of predicted and true days to 24 January

Trained on	Predicts	Level only	raw, station scale · LSTM	raw, station scale · Transformer	raw, day scale · LSTM	raw, day scale · Transformer
2015-16	**2018-19**	0.42	0.31 ± 0.12	0.60 ± 0.02	0.10 ± 0.01	0.22 ± 0.10
2018-19	**2015-16**	0.43	0.29 ± 0.04	0.60 ± 0.01	0.05 ± 0.11	−0.03 ± 0.03
2019-20	**2015-16**	0.43	0.51 ± 0.11	0.62 ± 0.01	0.12 ± 0.04	0.14 ± 0.04
2019-20	**2018-19**	0.42	0.46 ± 0.16	0.62 ± 0.00	0.08 ± 0.05	0.26 ± 0.06
2015-16	**2019-20 (earthquake)**	0.63	0.46 ± 0.07	0.68 ± 0.08	0.22 ± 0.03	0.02 ± 0.11
2018-19	**2019-20 (earthquake)**	0.63	0.41 ± 0.06	0.73 ± 0.02	0.19 ± 0.04	0.03 ± 0.02
2015-16 + 2019-20	**2018-19**	0.42	0.58 ± 0.03	0.61 ± 0.01	0.11 ± 0.09	0.49 ± 0.13
2018-19 + 2019-20	**2015-16**	0.43	0.53 ± 0.07	0.60 ± 0.01	0.11 ± 0.11	0.22 ± 0.07
2015-16 + 2018-19	**2019-20 (earthquake)**	0.63	0.48 ± 0.08	0.67 ± 0.03	0.18 ± 0.03	0.09 ± 0.06

Final-month AUC (days 0–30 against earlier days; 0.5 = chance)

Trained on	Predicts	Level only	raw, station scale · LSTM	raw, station scale · Transformer	raw, day scale · LSTM	raw, day scale · Transformer
2015-16	**2018-19**	0.77	0.85 ± 0.04	0.94 ± 0.01	0.80 ± 0.03	0.78 ± 0.11
2018-19	**2015-16**	0.72	0.70 ± 0.01	0.80 ± 0.03	0.63 ± 0.03	0.59 ± 0.05
2019-20	**2015-16**	0.72	0.75 ± 0.04	0.82 ± 0.01	0.59 ± 0.04	0.48 ± 0.02
2019-20	**2018-19**	0.77	0.87 ± 0.04	0.89 ± 0.01	0.81 ± 0.06	0.66 ± 0.05
2015-16	**2019-20 (earthquake)**	0.74	0.74 ± 0.04	0.87 ± 0.03	0.67 ± 0.01	0.58 ± 0.07
2018-19	**2019-20 (earthquake)**	0.74	0.71 ± 0.03	0.88 ± 0.01	0.67 ± 0.01	0.62 ± 0.01
2015-16 + 2019-20	**2018-19**	0.77	0.90 ± 0.02	0.89 ± 0.02	0.84 ± 0.01	0.74 ± 0.12
2018-19 + 2019-20	**2015-16**	0.72	0.72 ± 0.04	0.76 ± 0.01	0.57 ± 0.05	0.60 ± 0.05
2015-16 + 2018-19	**2019-20 (earthquake)**	0.74	0.74 ± 0.06	0.86 ± 0.02	0.66 ± 0.01	0.64 ± 0.04

MAE of log(1 + days)

Trained on	Predicts	Level only	raw, station scale · LSTM	raw, station scale · Transformer	raw, day scale · LSTM	raw, day scale · Transformer
2015-16	**2018-19**	0.56	0.38 ± 0.01	0.54 ± 0.06	0.38 ± 0.01	0.33 ± 0.01
2018-19	**2015-16**	0.82	0.82 ± 0.01	0.73 ± 0.00	0.82 ± 0.00	0.82 ± 0.00
2019-20	**2015-16**	0.96	0.80 ± 0.10	0.70 ± 0.01	0.91 ± 0.01	0.84 ± 0.01
2019-20	**2018-19**	0.59	0.41 ± 0.05	0.41 ± 0.02	0.45 ± 0.01	0.37 ± 0.01
2015-16	**2019-20 (earthquake)**	0.72	0.79 ± 0.01	0.62 ± 0.08	0.79 ± 0.01	0.81 ± 0.02
2018-19	**2019-20 (earthquake)**	0.79	0.76 ± 0.02	0.60 ± 0.02	0.78 ± 0.00	0.79 ± 0.01
2015-16 + 2019-20	**2018-19**	0.58	0.43 ± 0.05	0.50 ± 0.06	0.42 ± 0.02	0.48 ± 0.09
2018-19 + 2019-20	**2015-16**	0.88	0.75 ± 0.06	0.70 ± 0.01	0.84 ± 0.03	0.79 ± 0.02
2015-16 + 2018-19	**2019-20 (earthquake)**	0.75	0.73 ± 0.04	0.61 ± 0.02	0.77 ± 0.01	0.78 ± 0.02

Level only: a straight line on the day's level, fitted on the training winter and applied to the predicted one. A dash means that the predicted winter has fewer than five usable days in its last 30 days. The winters differ in their days (above), in particular 2018-19 has almost no day close to 24 January, so its rows are scored on earlier days only; the recordings of the three winters may also differ for reasons that have nothing to do with season or earthquake (instrument, site).

Cheat note: how to read the three tables

• A line is one pair: trained on the winter(s) of column 1, predicting the winter of column 2. The first six lines train on one winter, the last three on two. Cells are mean ± standard deviation over the seeds.

• Spearman is the main table; the final-month AUC and the MAE tell the same story. Compare a cell with the level only cell of its line.

• Lines predicting 2018-19: no day closer than 19 days to 24 January and 5 days in the last 30. Their low MAE is a narrow range of targets, and their AUC is rough.

• Station scale and day scale: see the note in section 9.

The placebo of section 9 at SVRC alone. The countdown is trained towards the earthquake and towards a day in January of the two other winters (2019-01-11, 2016-01-15; the closest days to 24 January that the recordings and the blocks allow), each with the 98 days before it, 4 folds by block age, 5 seeds. A fold has between about 20 and 60 training days here, so these numbers are rough; the scores are the mean over the seeds.

Event day	Validation days	of them 0–30 days	Level only MAE · AUC	raw, station scale · LSTM MAE · Spearman (folds) · AUC	raw, station scale · Transformer MAE · Spearman (folds) · AUC	raw, day scale · LSTM MAE · Spearman (folds) · AUC	raw, day scale · Transformer MAE · Spearman (folds) · AUC
2020-01-24 (earthquake)	53	19	0.92 · 0.44	0.74 · 0.24 · 0.70	0.65 · 0.59 · 0.85	0.74 · −0.17 · 0.65	0.75 · −0.02 · 0.65
2019-01-11	48	5	0.52 · 0.25	0.41 · 0.28 · 0.60	0.38 · 0.34 · 0.52	0.41 · 0.21 · 0.60	0.42 · 0.15 · 0.37
2016-01-15	34	7	0.81 · 0.39	0.55 · 0.47 · 0.64	0.60 · 0.23 · 0.65	0.55 · 0.33 · 0.61	0.60 · 0.02 · 0.57

Cheat note: how to read this table

- A line is one target date; a model cell is MAE · Spearman of the folds · final-month AUC.
- The earthquake has the highest Spearman of the three dates in 1 of 4 setups (raw, station scale · Transformer).
- Too little data to conclude from: about 20 to 60 training days per fold, and 5 and 7 days in the last 30 for the two other dates.

11. Pseudo-event placebo at a second earthquake: Balıkesir 2025

What was run. The placebo of section 9 at the Sındırgı (Balıkesir) mainshock of 10 August 2025, 16:53 UTC (Mw 6.1), which did not happen in winter: the day of the earthquake and 7 pseudo-events 14, 28, … days before it. The recordings of the stations PURS, STEP, DURS, BKES start in February 2025, so every event gets only the 84 days up to it (182 in section 9) and no earlier pseudo-event is possible. Raw signal, both scalings, both models, 4 folds, 3 seeds: 384 runs with 120–200 training station-days each; the predictions of the seeds are averaged day by day before scoring. The M6.1 of 27 October 2025 is not used as a second event: its history contains the August mainshock and its aftershocks.

Split. 14-day blocks counted back from the mainshock; 20 % of the 13 blocks are test blocks and were not read. For every event the other blocks are dealt into the folds in the order of their age, as in section 9.

Scores per event (out-of-fold predictions of its validation days): MAE of log(1 + days); the gain, i.e. the MAE of the level-only baseline minus the model's MAE; the Spearman correlation per fold, averaged; and the final-month AUC, the share of pairs of a day 0–30 days before the event and an earlier day in which the closer day gets the shorter prediction (0.5 = chance). Rank 1 is the best of the events. With 8 events the smallest share a rank can give is 1/8 = 0.12; the events share most of their days and are not independent.

The real event among the pseudo-events

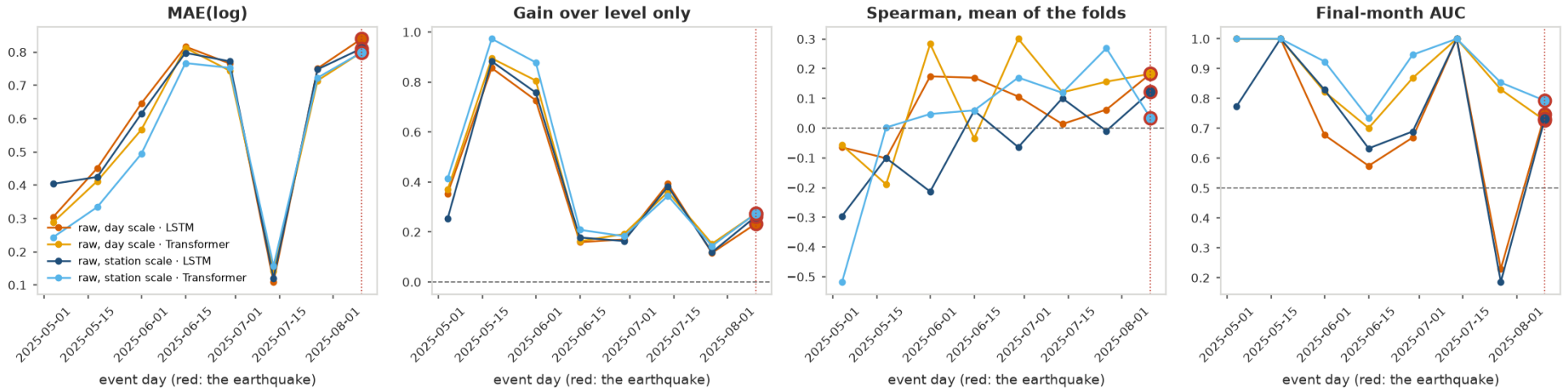
Input	Model	MAE(log) real · pseudo median (range) · rank of real	Gain over level only real · pseudo median (range) · rank of real	Spearman, mean of the folds real · pseudo median (range) · rank of real	Final-month AUC real · pseudo median (range) · rank of real
raw, station scale	LSTM	**0.81** · 0.61 (0.12 to 0.80) · **8 of 8**	**0.26** · 0.25 (0.12 to 0.88) · **4 of 8**	**0.12** · −0.06 (−0.30 to 0.10) · **1 of 8**	**0.73** · 0.77 (0.19 to 1.00) · **5 of 8**
raw, station scale	Transformer	**0.80** · 0.49 (0.16 to 0.77) · **8 of 8**	**0.27** · 0.34 (0.14 to 0.97) · **5 of 8**	**0.03** · 0.06 (−0.52 to 0.27) · **6 of 8**	**0.79** · 0.95 (0.73 to 1.00) · **7 of 8**
raw, station scale	level only	**1.07** · 0.94 (0.50 to 1.37) · **6 of 8**	–	–	**0.19** · 0.15 (0.00 to 0.30) · **4 of 8**
raw, day scale	LSTM	**0.84** · 0.65 (0.11 to 0.82) · **8 of 8**	**0.23** · 0.35 (0.12 to 0.86) · **5 of 8**	**0.18** · 0.06 (−0.10 to 0.17) · **1 of 8**	**0.75** · 0.68 (0.23 to 1.00) · **4 of 8**
raw, day scale	Transformer	**0.80** · 0.57 (0.14 to 0.81) · **7 of 8**	**0.27** · 0.36 (0.15 to 0.89) · **5 of 8**	**0.18** · 0.12 (−0.19 to 0.30) · **3 of 8**	**0.73** · 0.87 (0.70 to 1.00) · **7 of 8**
raw, day scale	level only	**1.07** · 0.94 (0.50 to 1.37) · **6 of 8**	–	–	**0.19** · 0.15 (0.00 to 0.30) · **4 of 8**

Cheat note: how to read the table above

• A cell is: real · pseudo median (range) · rank. Real is the score with the countdown aimed at the earthquake, pseudo the same score for the 7 dates without one. Rank 1 is the best; 1 of 8 is the most a rank can say.

• MAE: lower is better. Gain above 0: better than loudness alone. Spearman of the folds: are the days put in the right order (1 perfect, 0 none); the score to trust most. Final-month AUC: 0.5 is chance.

• The level only rows are loudness alone: a straight line on the level of the day.



Every event: MAE(log) · final-month AUC

Event day	Days before the earthquake	Validation station-days	of them 0–30 days	Level only MAE · AUC	raw, station scale · LSTM MAE · AUC	raw, station scale · Transformer MAE · AUC	raw, day scale · LSTM MAE · AUC	raw, day scale · Transformer MAE · AUC
2025-08-10 (earthquake)	0	184	76	1.07 · 0.19	0.81 · 0.73	0.80 · 0.79	0.84 · 0.75	0.80 · 0.73
2025-07-27	14	184	44	0.87 · 0.24	0.75 · 0.19	0.72 · 0.85	0.75 · 0.23	0.71 · 0.83
2025-07-13	28	128	32	0.50 · 0.10	0.12 · 1.00	0.16 · 1.00	0.11 · 1.00	0.14 · 1.00
2025-06-29	42	184	76	0.94 · 0.30	0.77 · 0.69	0.75 · 0.95	0.77 · 0.67	0.74 · 0.87
2025-06-15	56	184	76	0.98 · 0.24	0.80 · 0.63	0.77 · 0.73	0.82 · 0.57	0.81 · 0.70
2025-06-01	70	140	76	1.37 · 0.00	0.61 · 0.83	0.49 · 0.92	0.65 · 0.68	0.57 · 0.82
2025-05-18	84	152	44	1.31 · 0.00	0.42 · 1.00	0.33 · 1.00	0.45 · 1.00	0.41 · 1.00
2025-05-04	98	140	32	0.66 · 0.15	0.40 · 0.77	0.24 · 1.00	0.30 · 1.00	0.29 · 1.00

Notes. The station-days are those of the first row's input; both scalings use the same days. The events do not have the same validation days relative to themselves (columns 3 and 4), so the MAE of two events is not taken over the same range of times to the event. Events with fewer validation days in their last 30 days (2025-07-27, 2025-07-13, 2025-05-18, 2025-05-04) have a test block there, which is not read. The recording of the day of the earthquake is cut at 16:00 UTC, before the mainshock; the last day of a pseudo-event is a whole day. The epoch that is scored is chosen on the validation days, for every event in the same way. Commit recorded by the placebo runs: 25620fb0e8f3f77da816fcbc63a8b3c6729cf34c (uncommitted changes).

Cheat note: what the tables of this section show

• Spearman of the folds for the earthquake: 0.03 to 0.18 over the four setups; the highest of any pseudo-event: 0.30. For the two dates closest to the earthquake: −0.01 to 0.27 for 2025-07-27; 0.01 to 0.12 for 2025-07-13.

• Loudness alone, final-month AUC: 0.19 for the earthquake, median 0.15 for the pseudo-events.

• Table of every event: one line per target date, cells are MAE · final-month AUC. Compare a cell with the level only cell of its line, not with other lines: the lines differ in their validation days. A low MAE on a line with few days in the last 30 is a narrow range of targets, not a good prediction.

• Reading: no countdown to this earthquake. Where it ranks first on Spearman the value lies inside what the pseudo-events reach, and on MAE it is among the worst events. At Elazığ the Spearman of the earthquake was 0.58 to 0.79. Low power: 120 to 200 training station-days per run. Station scale and day scale: see the note in section 9.